F7AB - Actuarial and Financial Mathematics
Tutorial 3 – Nominal and Real interest rates

Question 1

1. Prove that
$$a_{\overline{n}|}^{(m)} = \frac{1 - v^n}{i^{(m)}}.$$
2. Hence show that
$$s_{\overline{n}|}^{(m)} = \frac{(1+i)^n - 1}{i^{(m)}}.$$

Question 2

1. State in words the meaning of the notation $\ddot{s}_{\overline{10}|}^{(12)}$.
2. State in words the meaning of the notation $600\ddot{a}_{\overline{10}|}^{(12)}$.
3. Prove that
$$\ddot{s}_{\overline{n}|}^{(m)i} = \frac{(1+i)^n - 1}{d^{(m)}}.$$
4. Hence show that
$$\ddot{a}_{\overline{n}|}^{(m)} = \frac{1 - v^n}{d^{(m)}}.$$

Question 3 Evaluate the following.

1. $s_{\overline{10}|}^{(2)\,0.05}$
2. $a_{\overline{17}|}^{(6)\,0.1}$
3. $\ddot{s}_{\overline{25}|}^{(4)\,0.035}$
4. $\ddot{a}_{\overline{7}|}^{(12)\,0.12}$
5. $\ddot{a}_{\overline{\infty}|}^{(4)\,0.06}$

Question 4 [Relationship between real and money yields (I).]

1. An investor pays £100 for an asset. Exactly one year later, the investor sells the asset for £110. Over the same year, the RPI has increased by the rate 3%.
   (a) Calculate the investor's money yield.
   (b) Calculate the investor's real yield.
2. An investor pays £1 for an asset. Exactly one year later, the investor sells the asset for £$(1+i)$, with $i > -1$. Over the same year, the RPI has increased at the rate q.
   Show that the real yield i' on the transaction satisfies
$$\boxed{1 + i' = \frac{1+i}{1+q}}$$

Question 5 [Relationship between real and money yields (II).] Assume that time is measured in years. At time 0, an investor pays £100 for an investment which produces a cashflow of £50 at time 1 and a cashflow of £60 at time 2. There are no further cashflows or value from the investment.

1. Calculate the investor's annualised money yield i.
2. Calculate the investor's annualised real yield i', based on the Inflation Index below.
3. Calculate the average annualised rate of change $\bar{q}$ in the Inflation Index.

Time	0	1	2
Inflation Index at time	100	104	112

4. Explain why, in general,
$$1 + i' \neq \frac{1+i}{1+\bar{q}}\,.$$

5. How must the inflation rate change as a function of time to have the equality $1+i' = \frac{1+i}{1+\bar{q}}$?

Question 6 A particular transaction will provide an investor with a yield of 6.400% per annum.

1. Assuming that over the period of the transaction the annual rate of inflation will be constant and equal to 2.800%, calculate the investor's real yield on the transaction.

2. Calculate the possible values of the constant annual rate of inflation q such that the investor obtains a real yield on the transaction of at least 3.750% per annum.

Question 7 A loan of £25 000 was issued in February 2010 and was repaid at par after three years. Interest accrues on the loan at the rate of 8% per annum and is payable annually in arrears.

The value of an Inflation Index at various times was as follows.

Date	Feb 2010	Feb 2011	Feb 2012	Feb 2013
Inflation index at date	205.0	215.6	223.5	231.5

Calculate the real rate of return on the loan by interpolating between two suitable interest rates which have a difference of at most 0.5%.

Answers: Q3.1. 12.73320; Q3.2. 8.34959; Q3.3. 39.79822; Q3.4. 4.85527; Q3.5. 17.28711: Q4.1(a). 10%; Q4.1(b). 6.796%; Q5.1. 6.394%; Q5.2. 1.077%; Q5.3. 5.830%; Q6.1. 3.502%; Q6.2. $q \leq 2.554\%$; Q7. 3.670%.

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Solutions to Tutorial 3 – Nominal and Real interest rates

Solution 1

1. Define $j := (1+i)^{1/m} - 1$. Then

$$a^{(m)i}_{\overline{n}|} = \frac{1}{m}\, a^{j}_{\overline{nm}|} = \frac{1}{m}\,\frac{1-(1+j)^{-nm}}{j} = \frac{1}{m}\,\frac{1-(1+i)^{-n}}{(1+i)^{1/m}-1}.$$

   As

$$1+i = \left(1+\frac{i^{(m)}}{m}\right)^m \quad \Rightarrow \quad i^{(m)} = m\cdot\left[(1+i)^{1/m}-1\right],$$

   then by substitution for $i^{(m)}$,

$$a^{(m)i}_{\overline{n}|} = \frac{1-(1+i)^{-n}}{i^{(m)}} = \frac{1-v^n}{i^{(m)}}.$$

2. As

$$s^{(m)}_{\overline{n}|} = (1+i)^n\, a^{(m)}_{\overline{n}|},$$

   it follows immediately from the first part that

$$s^{(m)}_{\overline{n}|} = \frac{(1+i)^n-1}{i^{(m)}}.$$

Solution 2

1. $\ddot{s}^{(12)}_{\overline{10}|}$ denotes the accumulation at the end of 10 years of a level annuity of £$\frac{1}{12}$ paid at the start of each month for 10 years.

2. $600\ddot{a}^{(12)}_{\overline{10}|}$ denotes the present value of a level annuity of £50 paid at the start of each month for 10 years.

3. Prove that

$$\ddot{s}^{(m)i}_{\overline{n}|} = \frac{(1+i)^n-1}{d^{(m)}}.$$

   Define $j := (1+i)^{1/m} - 1$. Then

$$\ddot{s}^{(m)i}_{\overline{n}|} = \frac{1}{m}\,\ddot{s}^{j}_{\overline{nm}|} = \frac{1}{m}\,\frac{(1+j)^{nm}-1}{\frac{j}{1+j}} = \frac{1}{m}\,\frac{(1+i)^n-1}{1-(1+i)^{-1/m}}.$$

   As

$$1-d = \left(1-\frac{d^{(m)}}{m}\right)^m \quad \Rightarrow \quad d^{(m)} = m\cdot\left[1-(1-d)^{1/m}\right] = m\cdot\left[1-(1+i)^{-1/m}\right],$$

   then by substitution for $d^{(m)}$,

$$\ddot{s}^{(m)i}_{\overline{n}|} = \frac{(1+i)^n-1}{d^{(m)}}.$$

4. As

$$\ddot{a}^{(m)}_{\overline{n}|} = v^n\,\ddot{s}^{(m)}_{\overline{n}|},$$

   it follows immediately from the first part that

$$\ddot{a}^{(m)}_{\overline{n}|} = \frac{1-v^n}{i^{(m)}}.$$

Solution 3 Either using the formula developed in the earlier questions or by changing the time unit:

1. $s^{(2)\,0.05}_{\overline{10}|} = \frac{1}{2}\, s^{0.024695}_{\overline{20}|} = 12.73320.$
2. $a^{(6)\,0.1}_{\overline{17}|} = \frac{1}{6}\, a^{0.016012}_{\overline{102}|} = 8.34959.$
3. $\ddot{s}^{(4)\,0.035}_{\overline{25}|} = \frac{1}{4}\, \ddot{s}^{0.008637}_{\overline{100}|} = 39.79822.$
4. $\ddot{a}^{(12)\,0.12}_{\overline{7}|} = \frac{1}{12}\, \ddot{a}^{0.009489}_{\overline{84}|} = 4.85527.$
5. As $v^n \to 0$ as $n \to \infty$,

$$\ddot{a}^{(m)}_{\overline{\infty}|} = \lim_{n\to\infty} \ddot{a}^{(m)}_{\overline{n}|} = \lim_{n\to\infty} \frac{1-v^n}{d^{(m)}} = \frac{1}{d^{(m)}} - \lim_{n\to\infty} \frac{v^n}{d^{(m)}} = \frac{1}{d^{(m)}},$$

we get $\ddot{a}^{(4)\,0.06}_{\overline{\infty}|} = 1/d^{(4)} = 17.28711$. Alternatively,

$$\ddot{a}^{(4)\,0.06}_{\overline{\infty}|} = \frac{1}{4}\, \ddot{a}^{0.014674}_{\overline{\infty}|} = 17.28711.$$

Solution 4

1. (a) The investor's money yield is $\frac{110}{100} - 1 = 10\%$.

   (b) Expressing the cashflow of £110 in terms of purchasing power at the start of the transaction year, its real value is $\frac{110}{1.03} = 106.796$. Using i' to denote the real yield, the equation of value is

$$100(1+i') - 106.796 = 0 \quad \Rightarrow \quad i' = \frac{106.796}{100} - 1 = 0.06796,$$

   i.e. the real yield is 6.796%.

2. Expressing the cashflow of £$(1+i)$ in terms of purchasing power at the start of the transaction year, its real value is $\frac{1+i}{1+q}$. Using i' to denote the real yield, the equation of value is

$$1 \times (1+i') - \frac{1+i}{1+q} = 0 \quad \Rightarrow \quad i' = \frac{1+i}{1+q} - 1.$$

Solution 5

1. The investor's money yield i is the solution to

$$100(1+i)^2 - 50(1+i) - 60 = 0. \tag{1}$$

   Solving as a quadratic equation, we find $i = 6.394\%$.

2. Expressing the cashflow of £50 at time 1 in terms of purchasing power at time 0, its real value is $\frac{50}{104} \times 100 = 48.077$. Expressing the cashflow of £60 at time 2 in terms of purchasing power at time 0, its real value is $\frac{60}{112} \times 100 = 53.571$.

   Using i' to denote the real yield, the equation of value is

$$100(1+i')^2 - 48.077(1+i') - 53.571 = 0. \tag{2}$$

   Solving as a quadratic equation, we find $i' = 1.077\%$.

3. The average rate of change $\bar{q}$ of the Inflation Index satisfies

$$1+\bar{q}=\left(\frac{112}{100}\right)^{1/2} \qquad \Rightarrow \qquad \bar{q}=0.05830.$$

4. For each money cashflow, we use the actual inflation rate at the time of the cashflow to convert it to a real value. Using the average inflation rate would result, in general, in a different value. Thus we cannot, in general, calculate the real yield directly from the money yield and the average inflation rate.

5. We have the equality $(1+i')=(1+i)/(1+\bar{q})$ when the inflation rate is constant across time, and/or in the simple case where we have only two time points at which cashflows occur (as in Question 5).

Solution 6 When the annual rate of inflation is constant, then the real yield i' and the money yield i are linked by the equation

$$1+i'=\frac{1+i}{1+q}, \tag{3}$$

in which q is the constant annual rate of inflation.

1. $i'=1.0640/1.0280-1=0.03502$, i.e. the real yield is 3.502% per annum.

2. We are asked to have $1+i'\geq 1.03750$. Using (3) and with $i>-1$ and $q>-1$, we find

$$\frac{1.0640}{1+q}\geq 1.03750 \quad \Rightarrow \quad q\leq 0.02554$$

   i.e. as long as the constant annual inflation rate is less than 2.554% per annum, the investor's real yield will exceed 3.75% per annum.

Solution 7

The amount of interest paid at end of each year is $0.08\times 25\,000=2\,000$. The real rate of return i' solves the equation of value

$$\begin{aligned} f(j) =& 25\,000(1+j)^3-2\,000\frac{205.0}{215.6}(1+j)^2-2\,000\frac{205.0}{223.5}(1+j)-27\,000\frac{205.0}{231.5} \\ =& 25\,000(1+j)^3-1\,901.67(1+j)^2-1\,834.45(1+j)-23\,909.29. \end{aligned}$$

at $f(i')=0$.

First estimate of i': the inflation index has increased over the three years by the factor $231.5/205=1.129$, which gives an average inflation rate of 4.1% p.a. As the money yield is 8% p.a. and the annual increases in the inflation index haven't been too volatile, then we would expect the real yield to be not too far from $\frac{1.08}{1.041}-1=3.75\%$ p.a.

By trial and error, we find that the solution i' lies between 3.5% and 4%. Then linear interpolation gives $i'\approx 0.03670$.

As a check, $f(0.03670)=-0.126$: acceptable given that we are working in £000s and not £00s.